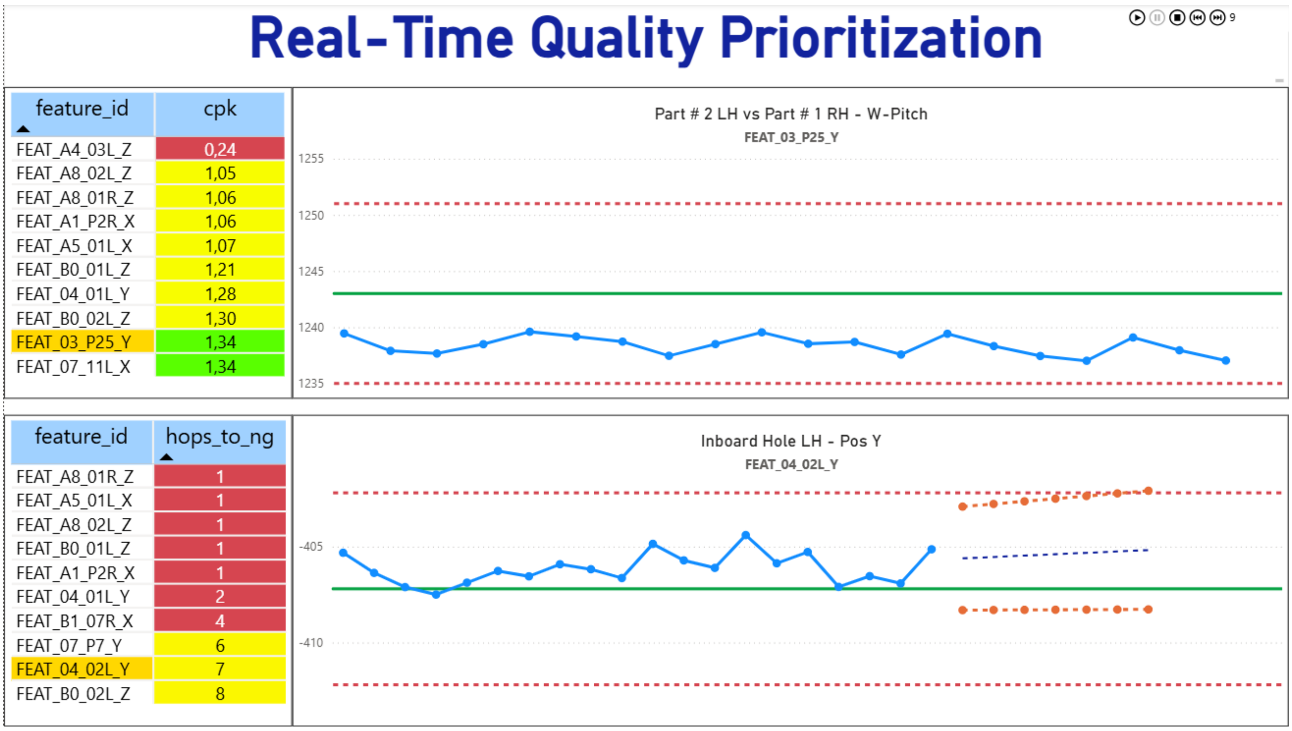
# Real-Time Quality Prioritization System

### Predictive Manufacturing Analytics for

### Process Capability & Quality Risk Monitoring




## Case Study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Industrial Data & Quality Systems


**Technologies:**

Python • DuckDB • Pandas • NumPy • SciPy • Power BI • Streamlit

*A real-time decision support system designed to transform dimensional measurement data into prioritized operational actions through statistical process capability analysis and predictive quality risk assessment.*

# The Challenge:

Modern manufacturing environments generate thousands of dimensional measurements every day. While Statistical Process Control (SPC) metrics such as **Cp** and **Cpk** provide valuable insight into process capability, quality engineers still face a fundamental operational challenge:

**Knowing where to focus their attention first.**

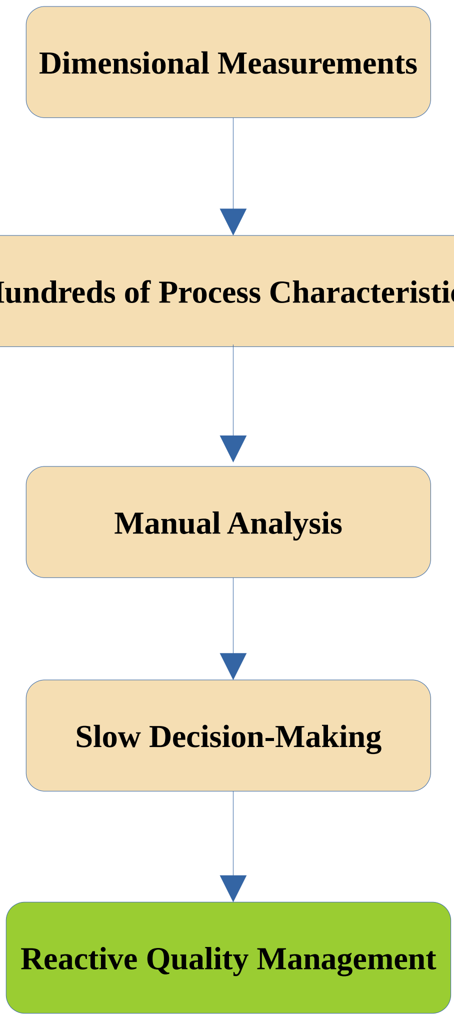
As the number of monitored process characteristics increases, manually reviewing capability reports becomes inefficient and reactive. Critical process deviations may remain unnoticed until product specifications are compromised, while engineers spend valuable time analyzing stable characteristics that require no immediate action.

Traditional reporting tools describe the current state of the process, but they rarely answer two operational questions:

- **Which process characteristics require immediate attention?**
- **Which characteristics are likely to become critical in the near future?**

To improve operational decision-making, manufacturing teams need a system capable of continuously prioritizing quality risks rather than simply reporting statistical indicators.

**Current scenario:**

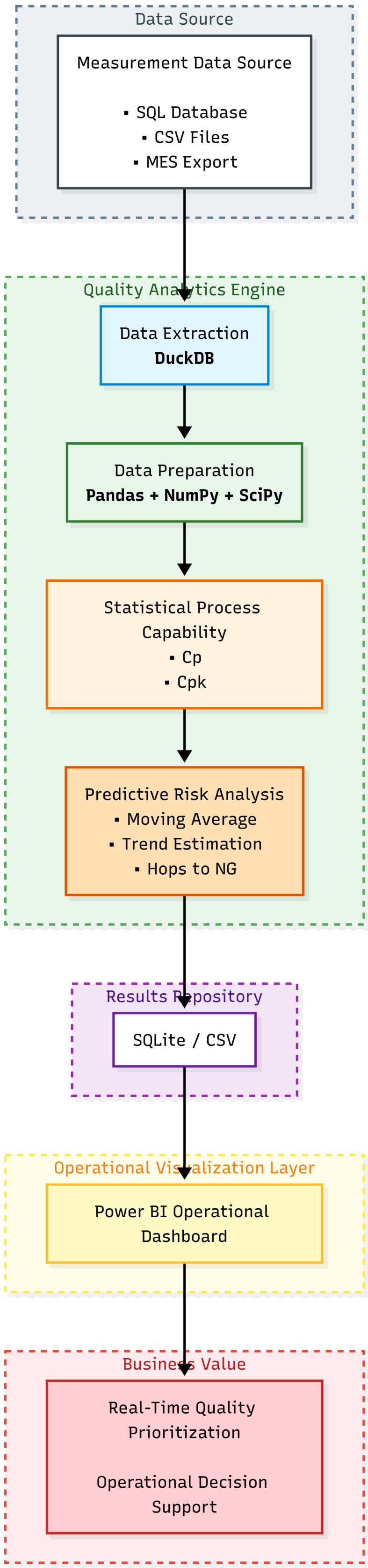


# Solution Overview:

To address this challenge, a complete analytics pipeline was designed to automatically transform raw dimensional measurement data into prioritized operational actions.

Instead of requiring engineers to manually inspect statistical reports, the system continuously analyzes process capability, predicts future process behavior, ranks process characteristics by operational priority, and presents the results through real-time operational screeners.

The solution was intentionally designed as a modular architecture, allowing each stage of the analytics pipeline to evolve independently while remaining compatible with existing manufacturing information systems.



# Quality Analytics Engine:

The Quality Analytics Engine transforms raw dimensional measurement data into prioritized operational insights through a combination of statistical process capability analysis and predictive quality assessment.

For each monitored process characteristic, the engine continuously evaluates both its current condition and its expected future behavior using the latest available production measurements.

The analytics workflow consists of two complementary stages:

## Current Process Capability

The engine calculates **Cp** and **Cpk** for every monitored feature, providing an objective assessment of the current process capability.

Process characteristics that already present insufficient capability are immediately classified as operational priorities.

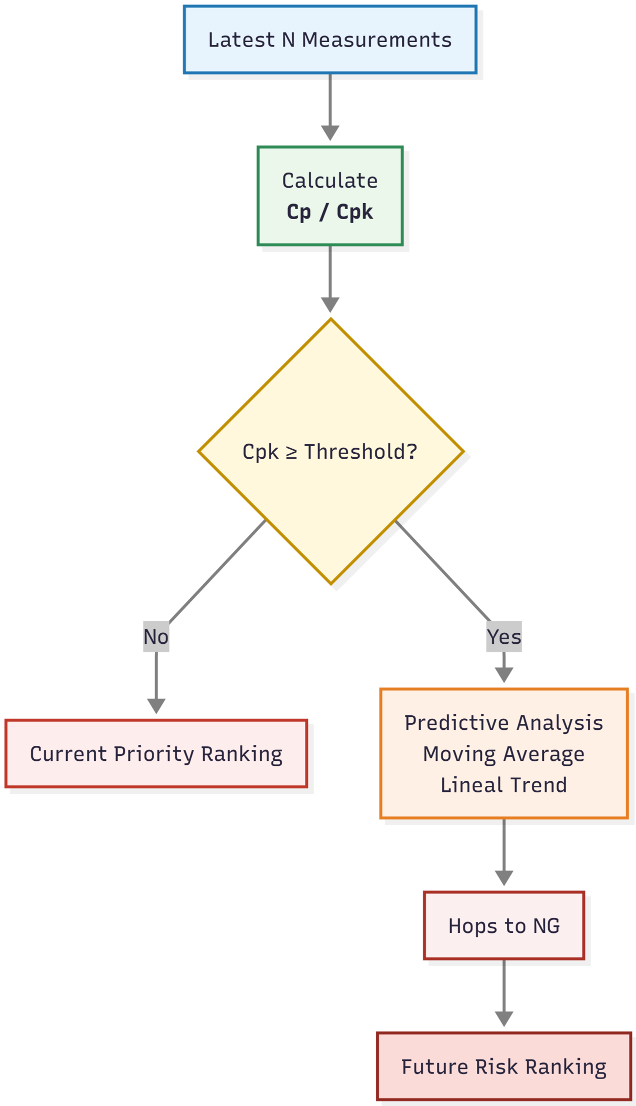
## Predictive Quality Risk

Process characteristics that remain statistically capable undergo a predictive analysis based on recent process behavior.

Using a moving average, lineal trend estimation and process variability, the engine estimates how many future measurements remain before the characteristic is expected to violate its specification limits.

This prediction is expressed as **Hops to NG**, a custom operational indicator representing the estimated number of individual measurements before a feature reaches a non-conforming condition.

By separating current capability assessment from predictive risk estimation, the system enables quality engineers to prioritize both immediate corrective actions and preventive interventions before customer specifications are exceeded.




## Key Innovation

### Hops to NG

***Hops to NG*** estimates the number of future individual measurements remaining before a process characteristic is expected to exceed its specification limits.

Unlike traditional capability indices, which describe the current condition of a manufacturing process, ***Hops to NG*** introduces a predictive perspective by estimating how many measurement cycles remain before customer specifications are expected to be violated.

This operational metric enables quality engineers to prioritize preventive actions before non-conforming parts are produced, shifting quality management from reactive problem-solving to proactive decision-making.

| Traditional SPC | Hops to NG |
|---|---|
| **How healthy is the process today?** | **How long until the process becomes critical?** |

# Operational Screeners:

## Current Process Capability

The Current Process Capability screener provides a real-time ranking of the process characteristics with the lowest **Cpk** values.

Its primary objective is to immediately identify manufacturing features that require corrective action due to insufficient process capability.

The screener continuously prioritizes the most critical process characteristics while simultaneously displaying the statistical history of the currently selected feature.

To support continuous monitoring in manufacturing environments, the dashboard automatically rotates through the highest-priority features without requiring operator interaction. This autonomous visualization mode ensures that unattended displays continuously present the most relevant quality information.

Only the highest-priority process characteristics are displayed at any given time, reducing cognitive overload and directing attention toward the operational decisions that matter most.

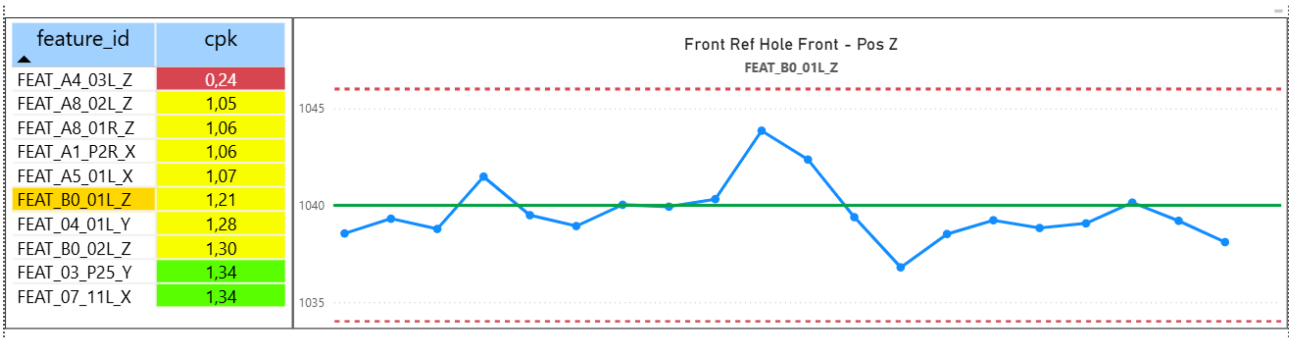



*Figure 1 - Current Process Capability Screener.*

**Purpose:**

• Prioritize current quality issues.

**Benefits:**

• Immediate operational visibility.

• Continuous unattended monitoring.

• Focus on the most critical process characteristics.

## Predictive Risk Screener:

The Predictive Risk screener focuses on process characteristics that are currently capable but exhibit statistical behavior suggesting an increased probability of future specification violations.

Instead of reacting to existing quality problems, this screener enables quality engineers to identify where preventive actions should be taken before customer specifications are exceeded.

Process characteristics are prioritized according to the **Hops to NG** metric, providing an operational estimate of how many future measurements remain before a feature is expected to become non-conforming.

Only the highest-priority process characteristics are displayed to minimize cognitive overload and direct attention toward the most critical operational decisions.

As with the capability screener, the dashboard automatically rotates through the highest-priority features, allowing unattended displays to continuously present the most urgent preventive actions.

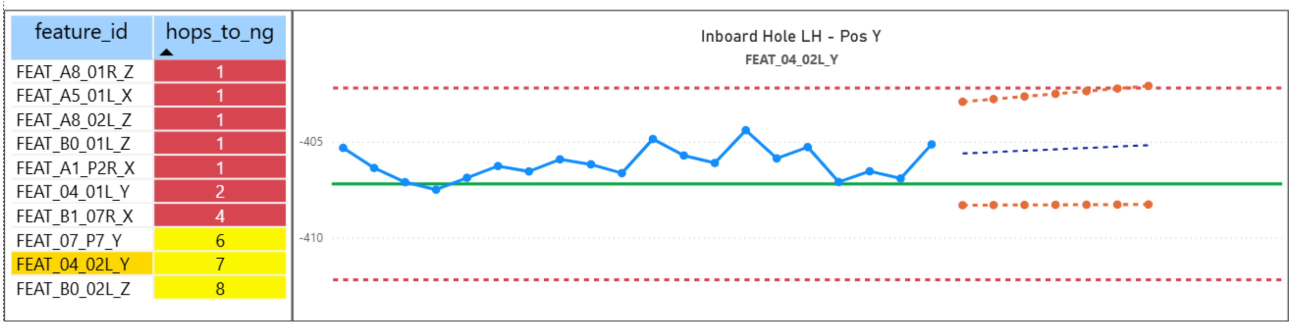



*Figure 2 – Predictive Risk Screener*

### Purpose:

• Prioritize preventive quality actions.

### Benefits:

• Early identification of future quality risks.

• Proactive decision-making.

• Reduced probability of producing non-conforming parts.

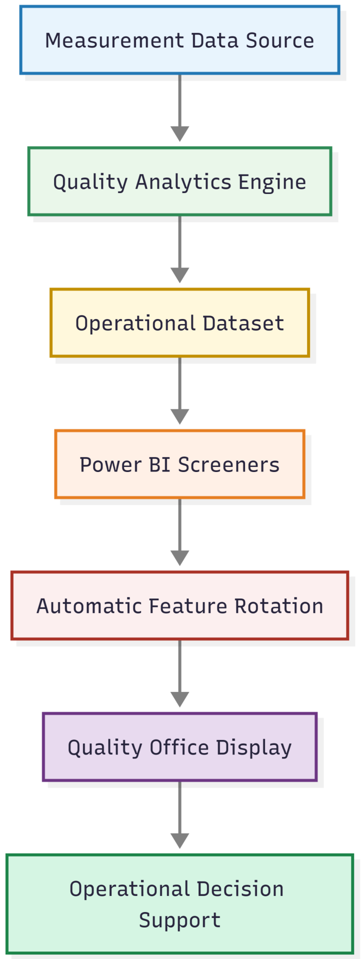
# Operational Deployment:

The system was designed as a continuously operating decision-support platform for manufacturing environments.

Instead of generating static reports for periodic review, the analytics engine executes automatically at configurable time intervals, processes the latest production measurements, updates the operational datasets, and refreshes the visualization layer without requiring user intervention.

The operational screeners are intended to run continuously on dedicated displays located in production or quality areas, providing manufacturing teams with an always-current view of both immediate process capability issues and emerging quality risks.

By combining autonomous data processing, automatic prioritization and unattended visualization, the system supports faster operational decisions while reducing the manual effort required to monitor hundreds of process characteristics.




**Designed for:**

√ *Continuous operation*

√ *Autonomous monitoring*

√ *Real-time prioritization*

√ *Unattended industrial displays*

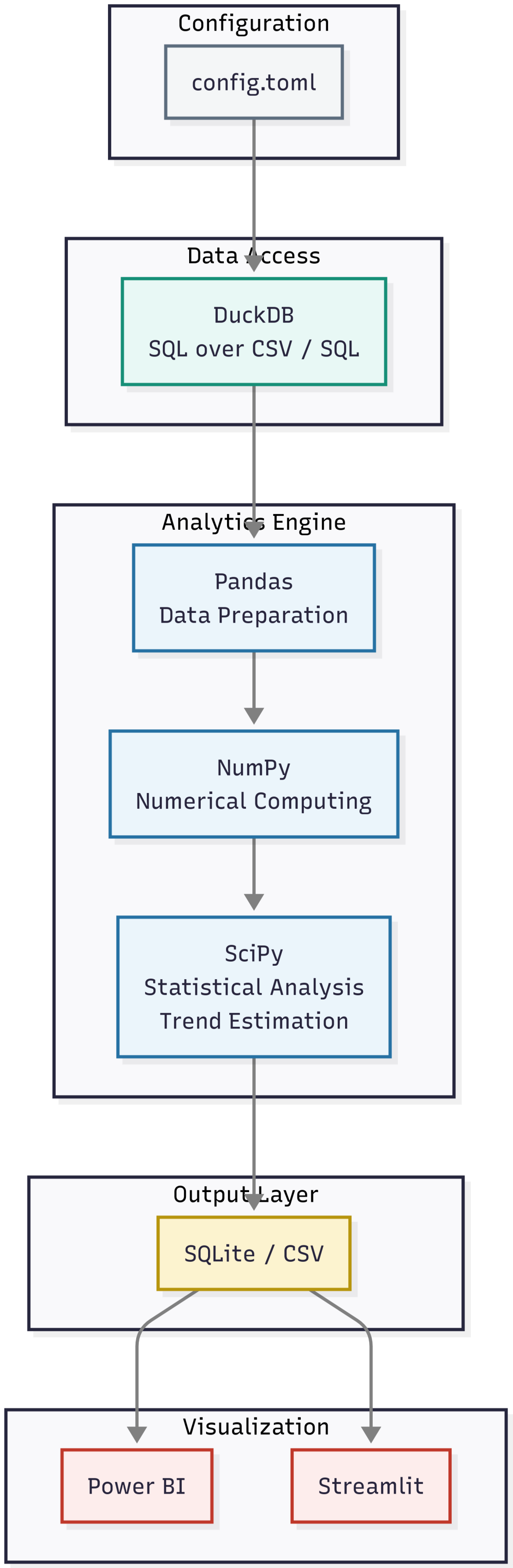
# Technical Stack

The solution was intentionally built using lightweight, open-source technologies to maximize portability, maintainability and ease of deployment in manufacturing environments.

Rather than relying on complex enterprise infrastructures, each component was selected according to its specific responsibility within the analytics pipeline.

| Component | Technology | Responsability |
|---|---|---|
| Data Extraction | DuckDB | Query measurement data directly from existing SQL databases and CSV files using standard SQL. |
| Data Processing | Pandas | Data manipulation, filtering and preparation for statistical analysis. |
| Numerical Computing | NumPy | High-performance numerical calculations. |
| Scientific Computing | SciPy | Trend estimation and predictive analysis. |
| Data Storage | SQLite / CSV | Portable output layer compatible with restricted IT environments. |
| Visualization | Power BI | Real-time operational screeners and decision support dashboards. |
| Configuration | TOML | Human-readable project configuration. |

The architecture remains modular, allowing individual components to be replaced without affecting the overall workflow.



# Engineering Decisions:

This project was developed under the same technical and operational constraints commonly found in industrial environments.

Rather than assuming ideal conditions, the solution was designed to integrate with existing manufacturing systems while respecting corporate IT policies and infrastructure limitations.

Key engineering decisions included:

- Reusing the existing production measurement database without modifying the legacy system.
- Supporting multiple output formats (SQLite and CSV) to maximize deployment flexibility.
- Building the analytics engine as an independent module, allowing future integration with different data sources.
- Separating the analytics engine from the visualization layer to improve maintainability.
- Using configurable parameters instead of hard-coded values to simplify future adaptations.

These decisions prioritize robustness, portability and maintainability over technological complexity, reflecting the practical requirements of industrial software development.

| Constraint | Engineering Decision |
|---|---|
| Legacy production system | Read-only integration |
| IT restrictions | SQLite / CSV support |
| Existing database | Non-invasive architecture |
| Future scalability | Modular analytics engine |
| Operational usability | Autonomous visualization |

# Results & Business Value:

The implementation of the Real-Time Quality Prioritization System introduces a shift in how manufacturing quality information is consumed and acted upon.

Instead of relying on periodic reports or manual inspection of statistical indicators, quality engineers are provided with a continuously updated prioritization of process characteristics based on both current capability and predictive risk.

This enables a transition from reactive quality management to a preventive and data-driven decision-making model.

## Operational Impact:

The system delivers three primary operational improvements:

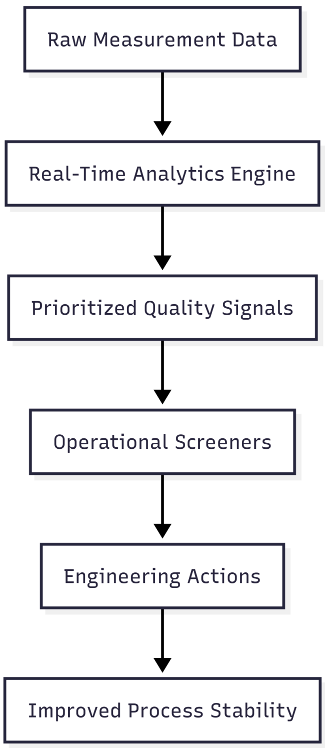
- Faster identification of critical process deviations through real-time capability monitoring.
- Early detection of potential quality risks using predictive analysis of process trends.
- Reduction in manual effort required to analyze and prioritize large volumes of measurement data.

## Strategic Impact:

Beyond operational efficiency, the system establishes a scalable framework for data-driven quality management.

By combining statistical process capability with predictive indicators such as Hops to NG, the solution enables organizations to allocate engineering resources based on operational urgency rather than static reporting metrics.

This approach improves both responsiveness and prioritization accuracy in manufacturing environments where multiple process characteristics must be monitored simultaneously.




Key Outcome: Shift from reactive quality control → predictive quality prioritization

# Conclusions & Future Work:

## Conclusions:

The Real-Time Quality Prioritization System demonstrates how manufacturing quality data can be transformed into an operational decision-support framework.

By combining statistical process capability analysis with predictive risk estimation, the system enables a transition from reactive quality monitoring to proactive prioritization of process characteristics.

The introduction of the **Hops to NG** metric provides a practical operational indicator that complements traditional capability indices, focusing on urgency rather than static process condition.

When deployed in a continuous monitoring environment, the system supports faster and more focused decision-making by automatically highlighting the most critical and most at-risk process characteristics.

## Future Work:

Several extensions can further enhance the capabilities of the system:

- Integration with live MES or SCADA systems for real-time data ingestion.
- Expansion of predictive models using machine learning techniques for improved accuracy of risk estimation.
- Incorporation of anomaly detection algorithms for early identification of abnormal process behavior.
- Deployment of alerting mechanisms to notify engineers when critical thresholds are reached.
- Extension of the system to multi-plant or supplier-level quality monitoring.

These improvements would allow the system to evolve from a local decision-support tool into a scalable enterprise-level quality intelligence platform.

# Appendix:

Github repository: https://github.com/juliancabmar/predictive-risk-screener

Operational Demonstration video: